February 24, 1992

Mr. Alfred Berkley
Alex. Brown & Sons
135 East Baltimore Street
Baltimore, MD 21202

Dear Mr. Berkley:

As a result of your request to review the technology of Sapiens International Corporations' Sapiens product, I visited the company on February 21st in Cary, NC. This letter is a report of my findings based on the visit and discussions with Sapiens International management.

In summary, there are no significant technical concerns regarding the product Sapiens or its underlying technology.

Technology Overview

The following points address the major areas of Sapiens technology:

- Robustness - Sapiens has the features needed for significant, real-world applications. It is capable of meeting the design and production requirements of a very large proportion of IBM mainframe applications. The ability of Sapiens to simultaneously manage multiple heterogeneous databases is the basis for distributed database computing -- an important industry trend -- and is not commonly found.

- Modular - Sapiens isolates programmers from IBM's systems software. This capability makes it easier for users to migrate applications to IBM's different systems environments, including MVS, VM, and VSE. It is ease of application migration which is at the heart of IBM's own still undelivered SAA architecture. This modularity should allow the company to readily adapt Sapiens to other computer environments.

- Object orientation - the Sapiens product encapsulates data and programming, and uses a message-passing architecture to communicate between objects. This is considered a modern -- and perhaps state-of-the-art -- architecture. Analysts generally consider the benefits of object orientation to include faster and more accurate development, easier maintenance, and flexible extensibility for applications -- all benefits needed today by mainstream mainframe developers. Further, Aberdeen firmly believes that today's distributed processing of transactions and queries by IS- and user-produced programs demands a strong automatic rulesbased processing capability. Sapiens has it.

I believe the architecture of Sapiens could have a decade-long useful life, and that there are no inherent obstacles to migrating the product to other hardware platforms.

- Productivity benefits - Sapiens claims advantages in application development and maintenance versus traditional 3GL programming languages (e.g. Cobol) and 4GLs (e.g. Nomad, Natural, and Ideal). I believe there are advantages, and that they mainly come from the code reusability of objects and from encapsulation.

There is no reason to disbelieve the competitive win by Sapiens in the Computerworld application development benchmark. However, the concepts of object-oriented application development are not well known by mainframe practitioners. It may take longer to train a Sapiens developer since the concepts of object-oriented programming as well as the specifics of Sapiens must be taught.

- Extensibility - By isolating database access routines from the core of Sapiens and from the application developer, Sapiens International can readily extend the product to other databases. Further, it should be easier for Sapiens International to add support for geographically distributed databases than competitors who have tightly bound their products to IBM's low-level mainframe data access methods.

The implication is that Sapiens could become an early market-supporter of distributed cooperative-processing applications, a market Aberdeen believes will grow exponentially during the 1990s.

- Positive Thinking - The need for developers to only have to program -- in many instances -- the positive application logic, and to leave to Sapiens the negative and reversal logic, is extremely powerful technology. I agree with the company's claims that typically about 20% of application coding is "positive" and the rest is "negative". The implication is that applications can require only 20% of traditional program logic while still properly handing error and reversal conditions. I probed in this area, and believe it is perhaps the most unique and differentiated area of Sapiens International's technology.

Positive Thinking is not a trivial feature. It is quite powerful and capable of resolving rather complex real-world transaction processing situations. My concern is that it will require developer training to anticipate the product's negative default actions, and that EDP auditors will demand to know what actions the "black box" portions of Sapiens are performing. In my opinion, the latter could be satisfied by product enhancements.

Performance

The Sapiens product is interpreted at runtime by an executive program. In general, interpreted code runs slower than equivalent code generated by a 3GL such as Cobol. In short, an interpreted implementation can trade-off fast development time for increased runtime computer overhead, which can also lead to increased user response times and a loss of user productivity.

The company claimed that the product performed with a 15%-30% performance penalty compared to a tuned 3GL application.

In my opinion, Sapiens' performance is adequate. This conclusion is based on the company's published 152-user benchmark results, and the fact that several customers have as many as 450 simultaneous active users. In my experience, few IBM mainframe applications require more than 250 users. In summary, I estimate Sapiens is applicable to over 80% of mainframe applications -- except where the greatest numbers of users are required or where absolute minimum response time speed is required. Page

Sapiens uses an IBM assembler-language-based runtime executive. It is unlikely that performance can be significantly improved from this base. However, any objectoriented development system has inherent overhead in message passing. In the future, straight-line-code competitors will have to add more relative overhead than will Sapiens in order to run in a distributed processing environment where message passing is mandatory.

Cooperative Processing

Aberdeen research shows a rapidly accelerating trend towards use of the client-server model, where PCs perform a significant part of application processing. The anticipated second quarter availability of a Microsoft Windows-based Sapiens client is important in order to satisfy immediate market demand, and to avoid competitive knockoffs from those who already have PC-client products.

I saw an off-line demonstration of the Windows product in pre-release form. The fact that IBM 3270-terminal-based Sapiens applications could automatically be switched to run as a Microsoft Windows client is a significant advantage.

Normal 3270 terminal screens compare poorly with the glitzy graphical user interface of Windows. In my opinion, Sapiens International has done a good job of using the graphical features of Windows given the 3270-screen limitations.

Downsizing
Another rapidly advancing industry trend observed by Aberdeen Group is the downsizing of new applications onto non-mainframe platforms. This threatens Sapiens International's potential market base.

The company indicated that an R&D project was underway to port Sapiens to Unix. The major technical obstacle is the requirement that the runtime kernel be rewritten from IBM assembler language to a Unix-common language such as C or C++. I believe this project is strategic because it will allow mainframe customers to see a migration path off their mainframes, if and when they choose to use it.

There is at present a need for cross-platform development tools which more closely cement IBM's mainframes with the RS/6000 Unix computers. There are no market leaders yet. Sapiens International has a market opportunity to follow IBM customers in their downsizing requirements, and another opportunity in integrating Unix systems with mainframe applications and databases.

Internal Database
Sapiens internally constructs a robust database management system (DBMS) on top of IBM's ubiquitous VSAM file system. The DBMS was at one time sold separately under the name DB1. There has been some concern among Sapiens International's investors that the Sapiens product is merely an enhanced DBMS.

While I cannot speak to the history of Sapiens, I do not share the concern. Sapiens needs the features of a DBMS in order to be able to robustly operate in a mainframe environment. VSAM alone is inadequate. Further, the DBMS footings of Sapiens are not discussed in the company's product marketing literature, and company management only brought up the topic after I did. Lastly, while the DB1 DBMS is bundled with the Sapiens product, buyers are free to use it or their own databases; that's one of the product's advantages.

Other Observations

o IBM relationship - The company clearly benefits from the joint marketing relationship with IBM. Based on my personal experiences at Stratus Computer (IBM System/88), Sapiens International is similarly benefiting from IBM's external sales contacts, from internal IBM interest, and from the halo effect that IBM's implied blessing confers on Sapiens International.

o Market Visibility - The company has no active program to reach industry influencers such as Aberdeen Group. Aberdeen is widely regarded as one of the top-five market researchers, yet we had nothing on Sapiens International in our extensive library. Neither have I seen any advertising.

o Demonstration and Presentation - The company's presentations and demonstrations must answer both technical and business benefit questions. I saw more emphasis on the former.

Conclusions

There are no significant technical concerns regarding Sapiens or its underlying technology. The company's technology can be differentiated from its competitors.

The company has a robust product which could appeal to a wide class of IBM mainframe buyers for more than 80% of commercial applications. The Sapiens product is applicable to both mission-critical transaction processing as well as querybased decision support. The modular architecture of the product lends itself to extensibility onto other IBM SAA platforms and into the open systems world. The technology architecture, with its emphasis on object-oriented design and development techniques, is early in its technology life cycle. The technology life should exceed the decade -- barring the unlikely total collapse of the mainframe market. Lastly, the Windows-based client product could answer buyer needs for client-server computing.

If you have any additional questions about my trip, observations, or Aberdeen's opinions on market conditions effecting Sapiens International, please do not hesitate to call.

Yours truly,

Peter S. Kastner
Vice President

2024 Metadata

Summary of Findings

The document is a report from Peter S. Kastner, Vice President of Aberdeen Group, to Mr. Alfred Berkley of Alex. Brown & Sons, dated February 24, 1992. The report details the findings from a visit to Sapiens International Corporation to review their Sapiens product. Key findings include:

- **Robustness**: Sapiens is capable of handling significant real-world applications and managing multiple heterogeneous databases, supporting distributed database computing.

- **Modularity**: The product isolates programmers from IBM's systems software, facilitating easier migration across different IBM environments.

- **Object Orientation**: Sapiens uses a modern architecture with encapsulated data and programming, offering benefits like faster development, easier maintenance, and flexible extensibility.

- **Productivity Benefits**: The product offers advantages in application development and maintenance over traditional 3GL and 4GL languages.

- **Extensibility**: Sapiens can be extended to other databases and supports geographically distributed databases.

- **Positive Thinking**: The product allows developers to focus on positive application logic, reducing the need for extensive negative and reversal logic coding.

- **Performance**: While interpreted code runs slower than compiled 3GL code, Sapiens' performance is deemed adequate for most IBM mainframe applications.

- **Cooperative Processing**: The upcoming Microsoft Windows-based Sapiens client is seen as a significant advantage.

- **Downsizing**: Sapiens is working on porting its product to Unix, addressing the trend of downsizing to non-mainframe platforms.

- **Internal Database**: Sapiens includes a robust DBMS built on IBM's VSAM file system, which is necessary for mainframe operations.

Recommendations for Computer Technology Buyers

Buyers should consider Sapiens for its robust, modular, and object-oriented architecture, which offers significant productivity and extensibility benefits. The product's ability to manage distributed databases and its upcoming Windows client make it a strong candidate for organizations looking to adopt client-server models. Additionally, the ongoing project to port Sapiens to Unix provides a future-proofing strategy for companies considering downsizing from mainframes.

Analysis of Technology in Relation to 2024

The technology discussed in the document, particularly the object-oriented architecture and modular design, aligns well with mainstream computing trends in 2024. Modern software development heavily relies on object-oriented principles and modularity to enhance scalability, maintainability, and integration capabilities. The emphasis on distributed database management and client-server models also reflects current trends in cloud computing and microservices architectures, which dominate the enterprise software landscape today.

Historical Impact on Companies

The document highlights Sapiens' early adoption of object-oriented and modular design principles, which positioned the company as a forward-thinking player in the software industry. This early focus on innovative architecture likely contributed to Sapiens' long-term success and its ability to adapt to evolving market demands. The company's strategic decisions, such as porting to Unix and developing a Windows client, demonstrated foresight in addressing industry trends, helping Sapiens maintain relevance and competitiveness over the decades.

Market Acceleration Catalysts

1. Adoption of object-oriented programming and modular design.

2. Support for distributed database computing.

3. Development of a Windows-based client for client-server models.

4. Porting to Unix to address downsizing trends.

5. Integration of robust DBMS capabilities.

6. Focus on productivity benefits through code reusability and encapsulation.

7. Positive Thinking feature reducing coding complexity.

8. Adequate performance for most mainframe applications.

9. Strong relationship with IBM for marketing and sales support.

10. Potential for early market leadership in distributed cooperative-processing applications.

Keywords

Robustness, Modularity, Object Orientation, Productivity, Extensibility, Performance, Cooperative Processing, Downsizing, Internal Database, Positive Thinking

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